

Revise

1. Integration of Spring MVC with hibernate.

Refer to flow diagram n code n steps

1.1 Explain what happens internally , when you start(deploy) Spring MVC Hibernate based web app on the web server ?

1.2 WC starts life cycle of D.S

1.3 In init() ->

Start the SC in web app , using hybrid approach

1.4 SC uses defaults for it's start up

name - servletName-servlet.xml (eg - spring-servlet.xml) => master config file

location - WEB-INF

contents - xml tags (enable class internal annotation , base-pkg for comp scan,enable mvc annotation support)

configure ViewResolver

import - hibernate-persistence.xml

DBCP bean , LocalSessionFactory bean(provider of SF) , Tx manager

1.5 SC applies reflection - starts scanning - base pkg

Controller - Service -DAO

2. Understand the detailed configuration of - hibernate-persistence.xml

2.1 Configure a spring bean , Apache supplied Basic Data Source that represents DBCP.

Props - driver class , db url , user name ,password , init size of pool n max size of pool.

2.2 Configure LocalSessionFactory as a spring bean , to provide SessionFactory , that can be autowired n injected in DAO layer.

Props - packagesToScan : to scan the entities package.

Depcy - data source bean created in 2.1

2.3 Configure HibernateTransactionManager as a spring bean.

Depcy - LocalSessionFactoryBean

Why - To automatically manage transactions (resulting in reduced boilerplate code in DAO layer)

2.4 Enable annotation support for txs

(i.e manage txs using @Transactional)

JSTL Action for URL rewriting + generating URLs relative to current ctx path

<c:url var="attrName" value="url"/>

3. Continue with earlier project n solve below objectives.

drop database blogs

use sample data to populate - users , categories n posts

use day9.1 , to populate data

RegisterUser , Add Category , Add post , Add Comment

1. After client sends a request to the above web app -

Request response flow

1.1 clnt sends request

http://host:port/ctx\_path/ , method=GET

Resp - index.jsp

## 1.2 clnt clicks on the link "User Login"

Request URL - [http://localhost:8080/ctx\\_path/user/login](http://localhost:8080/ctx_path/user/login) , method=GET

OR appended by jsessionid (in case of blocked cookies)

Resp - login.jsp

## 1.3 clnt submits the form to the same URL , with POST

Request URL - [http://localhost:8080/ctx\\_path/user/login](http://localhost:8080/ctx_path/user/login) , method=POST

Process the form - authenticate blogger - forward blogger to home page , displaying blogger details.

## 4. Understood the efforts of the extensive configuration ?

Any solution : YES ,

## 5. Enter Spring Boot

Readmes to refer

1."day12-data\day12\_help\spring boot\spring-boot-steps.txt"

2. "day12-data\day12\_help\spring boot\Why n What is spring boot.txt"

## 6. Port existing web application with spring boot

## 7. While Porting earlier app to spring boot

Any problems noticed ?

Exception : org.hibernate.HibernateException

Root cause : no curnt session context configured!

(i.e unable to get curnt session from SessionFactory , in the DAO layer)

Reason : Spring boot DOES NOT directly support native hibernate APIs

(eg : org.hibernate : SessionFactory , Session, Transaction....)

Soln : Replace Hibernate by it's specs : JPA (Java Persistence API)

## 8. Replace hibernate native APIs by JPA

Refer : "day12-data\day12\_help\spring boot\diagrams\jpa-entitymgr-session-layers.png"

## 9. BUT if you notice , in all DAO layers (JDBC / Hibernate / JPA) , you are solving common problems :

CRUD

So is there any out of box (ready made) solution ?

YES

## 10. Enter Spring Data JPA n use it in the existing spring boot web MVC app.

Refer : "day12-data\day12\_help\Spring Data JPA\regarding spring data JPA.txt"

Spring Data JPA Links (Bookmark them!)

1. API Docs : <https://docs.spring.io/spring-data/jpa/docs/2.7.18/api/>

2. Ref docs : <https://docs.spring.io/spring-data/jpa/docs/2.7.18/reference/html/>

## 11. Use lombok library , to reduce boilerplate code.

Refer - "day12-data\day12\_help\spring boot\Regarding Lombok.txt"

## 12. Solve Blogs assignment from login -logout

Refer - "day12-data\day12\_help\blogs related\blogs assignment.jpg"

## 12.1 Problem observed with default navigation technique(i.e RD's forward)

Multiple submission issue.

Solution -Use PRG (post-redirect-get) pattern.

Replace forward by redirect.

HOW ?

eg : return "redirect:/blogger/home";

12.2 To store the attributes under current request scope - Model map

To store the attributes under current session scope - HttpSession

To store the attributes till next redirect request only , use RedirectAttributes

SC can inject RedirectAttributes directly in req handling method , as a dependency.

Use it's API

public RedirectAttributes addFlashAttribute(String attrName , Object value)

Regarding java.util.Optional Class

To throw custom exception , use

public T orElseThrow(Supplier excSupplier)

Returns a value(T) if present , otherwise throws specified exception

eg : Suppose AuthenticationException : custom unchecked exception

userDao.findByEmailAndPassword(em,pass).orElseThrow(() -> new  
AuthenticationException("invalid email or pwd"));